

Hierwave: the mathematics

Companion to `notes/reference.md` (October 2026)

The models, the sampler, and the training algorithm, as equations. Definitions only; the arguments are in `channels.tex` and the per-experiment notes. Section numbers follow `reference.md`.

The setting: a world is an $H \times W$ grid of *tiles* (voxels in 3D) described at several *levels* of resolution. A level is a block size $h \in \{1, 2, 4, \dots\}$; level h partitions the tile grid into $h \times h$ blocks and holds one value per block. The levels together form a Markov random field (MRF) whose factors are small tables; generation runs top-down, each level sampled with the coarser ones held fixed.

1 Objects

Channels and views. A channel c at block size h_c holds a state $z_c : \Lambda_c \rightarrow \mathcal{D}_c = \{0, \dots, D_c - 1\}$ on the grid $\Lambda_c = \{0, \dots, H/h_c - 1\} \times \{0, \dots, W/h_c - 1\}$; D_c is its number of values. The tile channel has $h = 1$. A *view* of c is a map $\alpha : \mathcal{D}_c \rightarrow \{0, \dots, V_\alpha - 1\}$ onto a small feature; factors read views, never raw values. A *fixed* mask $\phi_c \subseteq \Lambda_c$ marks cells the sampler never updates.

Parent cell and block. For $p \in \Lambda_a$ and a coarser-or-equal channel b ($h_a \leq h_b$), the cell of b containing p , shifted by an offset δ in b 's cells, is

$$q_b(p, \delta) = \left\lfloor \frac{p h_a}{h_b} \right\rfloor + \delta, \quad \delta \in \mathbb{Z}^2; \quad (1)$$

conversely $\text{block}(q) = \{p \in \Lambda_a : q_b(p, 0) = q\}$ is the set of a -cells inside the b -cell q .

Factors. A factor is anything that assigns each candidate value at a cell a local energy. Three kinds exist: tables (this paragraph), the conv potential (Section 5), and the certificate (below), whose energy is computed by a procedure. Tables are the fast path (a candidate's energy is a lookup) and are also what training produces. A table factor is a table T in energy units (nats), with $+\infty$ for hard rules, indexed by views of one or two channels. Writing a for the finer (or only) channel and b for the other, the three kinds are

$$\text{pair: } E_{a\alpha, b\beta, \delta}^{\text{pair}}(z) = \sum_{p \in \Lambda_a} T[\alpha(z_a(p)), \beta^{\text{pad}}(z_b(q_b(p, \delta)))], \quad (2)$$

$$\text{count: } E_{a\alpha, b\beta}^{\text{count}}(z) = \sum_{q \in \Lambda_b} T\left[\beta(z_b(q)), \sum_{p \in \text{block}(q)} \alpha(z_a(p))\right], \quad h_a < h_b, \quad (3)$$

$$\text{unary: } E_{a\alpha}^{\text{un}}(z) = \sum_{p \in \Lambda_a} T[\alpha(z_a(p))]. \quad (4)$$

A pair factor is one offset δ applied at every cell of a , i.e. a translation-invariant term with a position-free table; a neighbourhood is a set of such factors, one per offset, and the largest $|\delta|$

among a channel's factors sets its sampler's radius. With $b = a$ and $\delta \neq 0$ it is a same-level neighbour term; with $h_b > h_a$ it is a parent read (the parent cell, or a parent cell δ away). β^{pad} returns the factor's pad value when $q_b(p, \delta)$ is off the grid, and the term is dropped when the pad is -1 . A factor is *homed* on its finer side a . The sampler of a channel reads only the factors homed on it, plus the reflections of same-level pairs (transposed table, negated offset), so during generation a coarse channel never sees a finer one (top-down).

Honour factors. A factor is *honour* when it makes a finer channel agree with a view of a coarser one, i.e. a pair factor whose table penalises $\alpha(z_a(p)) \neq \beta(z_b(q_b(p, 0)))$: hard ($+\infty$) for guarantees, soft (a cost κ per violated cell) otherwise. A coarse value is *honourable* if some finer completion has finite energy.

Certificate. The factor that enforces connectivity or support. It lives on a tile channel with three views, $m = \text{mass}$ (0: not part of the structure), $\tau = \text{trunk}$ (1: a root of the structure), and *join bits* $J_d(t, t') \in \{0, 1\}$ saying whether tile value t joins value t' across side $d \in \{\text{N, E, S, W}\}$; and on a companion distance channel $d : \Lambda \rightarrow \{0, \dots, D_{\max}, \text{INF}\}$ at the same level, $\text{INF} = D_{\max} + 1$. Write $t_p = z(p)$, $q \sim_d p$ for the 4-neighbour of p across side d . Cell p is *valid* iff

$$(m_p = 0 \wedge d_p = \text{INF}) \vee (\tau_p = 1 \wedge d_p = 0) \vee (d_p < \text{INF} \wedge \exists q \sim_d p : J_d(t_p, t_q) \wedge m_q \geq m_p \wedge d_q < d_p). \quad (5)$$

A neighbour q satisfying the last clause is a *witness* for p ; p 's *dependants* are the neighbours for which p is the only witness. The energy is $+\infty$ on an invalid cell and γd_p otherwise (γ is **delta** in the code). With the **tree** option, exactly one neighbour may witness p (its parent) and every other joined neighbour must be a child ($m_q \leq m_p$, $d_q > d_p$), so the join graph is a forest.

Painted channels. A *painter* $\pi_c : \mathcal{D}_c^{\Lambda_c} \rightarrow \mathcal{D}_w^{\Lambda_w}$, $h_w < h_c$, writes a finer channel w deterministically from z_c ; w is fixed everywhere and is rewritten whenever z_c changes. Honour factors then read w through position-free pair tables; the dependence on where a cell sits inside its block (which tiles of a block are in an object's disc, say) lives in π_c . The *spill* of a painter is the set of w -cells outside the blocks of a region that the painter writes from values inside it.

Promotion. A hard factor on a view α of a channel u with many values (an exemplar coordinate, say) rejects most of u 's candidates and can leave no candidate at all; promotion moves the hard factor onto a small channel that is updated exactly, and lets u pay only a soft cost for disagreeing with it. Add a channel w at the same level with $\mathcal{D}_w = \text{range}(\alpha)$, move the factors that read $\alpha(z_u)$ to read w , and add the cell-wise factor

$$E^{\text{cell}}(w_p, u_p) = \text{cost}(w_p, \alpha(u_p)), \quad (6)$$

with cost one of: $\nu \mathbb{1}[w \neq \alpha(u)]$ (a soft copy, which tempers the hard factors that now read w); $-\log \hat{p}(w | u)$ counted from an exemplar; or the induced free energy of Section 2. Marginalising w recovers the original model with its hard factors softened by exactly cost.

2 The two models

Index the levels $\ell = L, \dots, 1$ from coarsest to finest, block sizes $h_L > \dots > h_1 = 1$. Let z_ℓ be the channels at level ℓ , $z_{>\ell}$ the coarser ones, $z_{\geq \ell}$ both, and $E_\ell(z_\ell | z_{>\ell})$ the sum of the factors homed on level ℓ (they read level ℓ and coarser levels only).

Bidirectional (the ideal). The undirected joint of every declared factor,

$$p^*(z) = \frac{1}{Z} \exp \left(- \sum_{\ell} E_{\ell}(z_{\ell} \mid z_{>\ell}) \right). \quad (7)$$

It is the specification; it is never sampled in production.

Forward (the chain, what ships).

$$q(z) = \prod_{\ell=L}^1 q_{\ell}(z_{\ell} \mid z_{>\ell}), \quad (8)$$

where q_{ℓ} is the law of level ℓ after S_{ℓ} sweeps of a kernel that leaves $\exp(-(E_{\ell} + \Phi_{\ell}))$ invariant (Gibbs for small domains, candidate-set Gibbs for large ones; Section 3) at temperature $T = 1$, started from the *painted initialisation* (a consistent refinement of $z_{>\ell}$ written by the painters), with $z_{>\ell}$ clamped. $\Phi_{\ell}(z_{\ell} \mid z_{>\ell})$ is the *learned potential* of level ℓ . We call S_{ℓ} sweeps of this kernel the *budgeted kernel*.

The identity. Define the free energy of the levels below ℓ , recursively from the bottom,

$$F_{\ell}(z_{>\ell}) = -\log \sum_{z_{\ell}} \exp \left(-E_{\ell}(z_{\ell} \mid z_{>\ell}) - F_{\ell-1}(z_{\geq \ell}) \right), \quad F_0 \equiv 0. \quad (9)$$

Then $p^*(z_{\geq \ell}) \propto \exp \left(-\sum_{\ell' \geq \ell} E_{\ell'} - F_{\ell-1}(z_{\geq \ell}) \right)$, and the chain with equilibrated kernels reproduces p^* exactly iff

$$\Phi_{\ell} = F_{\ell-1} \quad \text{for every level } \ell. \quad (10)$$

The learned potential of a level is the free energy the levels below induce on it. With the budgeted kernel in place of the exact sum in the definition of F , the identity holds for the chain *as it runs* (the free energy as that kernel realises it at that budget).

Cluster expansion. $F_{\ell-1}$ is approximated by local terms over the cells of level ℓ ,

$$F_{\ell-1}(z_{\ell}) \approx \sum_p f_1(z_p) + \sum_{\langle p,q \rangle} f_2(z_p, z_q) + \cdots, \quad (11)$$

exact in the limit where the correlation length of the finer levels is below the block size (off-critical content); the residual of the fit (Section 4) measures the omitted orders.

Support and finite part. The support of Φ_{ℓ} ($+\infty$ on coarse values that are not honourable) comes from honourability checks, not from the fit; only the finite part is learned.

3 The sampler

A sweep of a channel at level ℓ visits every non-fixed site p once. At p , with everything else fixed, the candidate energy $e_p(t)$ is the sum of the factors homed on the channel that involve p , evaluated with $z_p = t$: the designed terms of E_{ℓ} (reading level ℓ and coarser) and the tables of Φ_{ℓ} , which is a

same-level potential and enters as ordinary unary and pair rows. Factors homed on finer levels are never read; Φ_ℓ stands in for them. The site draws its exact conditional,

$$t \sim \frac{\exp(-e_p(t)/T)}{\sum_{t'} \exp(-e_p(t')/T)} \quad \text{as} \quad \arg \max_{t: e_p(t) < \infty} [-e_p(t)/T + G_t], \quad G_t \sim \text{Gumbel}(0, 1); \quad (12)$$

a site with no finite candidate keeps its value. Sites are visited colour by colour so that sites of one colour do not interact: $(x + y) \bmod 2$ when every factor has radius 1, $(x + 2y) \bmod 5$ at radius 2. With a certificate the draw is joint in (t, d) : for each t the valid d form an interval $[a_t, b_t]$ (a_t one above the smallest witness d , b_t the smallest d among p 's dependants, which must stay valid), so

$$w_t = \exp(-e_p(t)/T) \sum_{d=a_t}^{b_t-1} e^{-\gamma d/T} = \exp(-e_p(t)/T) e^{-\gamma a_t/T} \frac{1 - e^{-\gamma(b_t-a_t)/T}}{1 - e^{-\gamma/T}}, \quad (13)$$

t by Gumbel-max over $\log w_t$, then d from the truncated geometric law on $[a_t, b_t]$. Dependants read their other neighbours, hence radius 2.

Large domains. Enumerating $\mathcal{D}_c$ at every site is only for small D_c . A channel with a large domain (an exemplar coordinate, D_c = the exemplar's cells) draws over a *candidate set* $C_p \subset \mathcal{D}_c$ instead: the current value, the coherent continuation $z_q + (p - q)$ of each neighbour q , K random values, and K_t random values consistent with the tile under p . The draw is the same Gumbel-max restricted to C_p . Because C_p contains the current value and is chosen without reference to z_p , the step leaves the level's conditional invariant, which is all the chain needs. Cost per site: $|C_p|$ energy evaluations, each over the candidate's footprint; Φ_ℓ is then evaluated on the fly on that footprint rather than read from tables. Any such invariant kernel may stand in for the Gibbs sweep in q_ℓ and in the identity of Section 2.

4 The training (combining) algorithm

Input: a coarse channel c at level ℓ ; the levels below it, already corrected (carrying $E + \Phi$); its painter π ; and a *reference value* $r \in \mathcal{D}_c$, the background value (empty, absent) against which every other value is measured. Output: Φ_ℓ as tables for the forward kernel. Nothing samples c itself.

Windows. A window $W \subset \Lambda_c$ (e.g. 3×3 cells) with halo ∂W (the one-cell ring around it). Draw z_W and $z_{\partial W}$ from a proposal (uniform at first, later windows cut from forward samples); set c to r outside $W \cup \partial W$. Let z_W^r be the configuration with r on all of W and the same halo. The fine region $R(W)$ is the blocks of $W \cup \partial W$ plus the painters' spill.

Free energy per window. With E_{fine} the energy of the levels below c restricted to $R(W)$ (the coarse values clamped),

$$F(z_W \mid z_{\partial W}) = -\log \sum_{z_{R(W)}} \exp(-E_{\text{fine}}(z_{R(W)} \mid z_W, z_{\partial W})), \quad y(W) = F(z_W \mid z_{\partial W}) - F(z_W^r \mid z_{\partial W}). \quad (14)$$

F is computed exactly where the toy allows it (transfer matrix; per-cell products when the fine cells are independent given the paint), otherwise by annealed importance sampling (AIS) with the forward model's own fine kernel. For AIS split $E_{\text{fine}} = E_{\text{un}} + E_{\text{rest}}$ (the unary factors of the fine

channel vs. everything else), replace $+\infty$ by a finite Λ (30) in E_{rest} , and take inverse temperatures $\beta_0 = 10^{-3} < \beta_1 < \dots < \beta_K = 1$ (linear). With $E_{\text{un},s}(t)$ the unary energy of value t at fine site s ,

$$\log Z_0 = \sum_{s \in R(W)} \log \sum_t e^{-E_{\text{un},s}(t)}, \quad (15)$$

$$x_0 \sim p_0 \propto e^{-E_{\text{un}}}, \quad x_k \sim \text{one Gibbs sweep of } p_{\beta_k} \propto e^{-E_{\text{un}} - \beta_k E_{\text{rest}}} \text{ from } x_{k-1}, \quad (16)$$

$$\log w = - \sum_{k=0}^{K-1} (\beta_{k+1} - \beta_k) E_{\text{rest}}(x_k), \quad (17)$$

$$\widehat{\log Z} = \log Z_0 + \text{logmeanexp}_{m=1..M} \log w^{(m)}, \quad \hat{F} = -\widehat{\log Z}, \quad (18)$$

over M independent chains. $\hat{F}$ is biased upward at small K ; $K = 256$, $M = 4$ – 8 was adequate on the toys.

Paint potential. The parametrisation of Φ_ℓ that generalises to unseen value pairs: a local energy over the painted image rather than over c 's values. Let $\pi = \pi(z_c)$ be the painted channel at fine resolution with values $\{0, \dots, V-1\}$, indexed by fine cells t , and $\mathcal{O} = \{(0, 1), (1, 0), (1, 1), (1, -1)\}$ the four independent offsets of the 8-neighbourhood. Then

$$F_\theta(\pi) = \sum_t u(\pi_t) + \sum_{\delta \in \mathcal{O}} \sum_t g_\delta(\pi_t, \pi_{t+\delta}), \quad \theta = (u, \{g_\delta\}), \quad (19)$$

off-grid partners dropped. $F_\theta = N(\pi) \cdot \theta$ is linear in θ through the count features $N(\pi) \in \mathbb{R}^{V+|\mathcal{O}|V^2}$ (how often each value, and each offset pair of values, occurs). Regression with the halo painted:

$$x(W) = N(\pi(z_W, z_{\partial W})) - N(\pi(z_W^r, z_{\partial W})), \quad \hat{\theta} = \arg \min_{\theta} \sum_W (x(W) \cdot \theta - y(W))^2 + \lambda \|\theta\|^2. \quad (20)$$

The halo's own terms cancel in x and y ; the window-halo cross terms are features. The relative residual $\sqrt{\sum_W (x \cdot \hat{\theta} - y)^2 / \sum_W y^2}$ measures what is not local in the painted map at this order, down to the AIS noise floor.

Materialisation. Tables over $\mathcal{D}_c$ for the forward kernel. Write $\pi(v)$ for the paint of a configuration with v at one cell and r everywhere else, and $\pi(v \text{ at } p, v' \text{ at } p+d)$ likewise for a pair. Then

$$u_c(v) = F_{\hat{\theta}}(\pi(v)) - F_{\hat{\theta}}(\pi(r)), \quad (21)$$

$$g_{c,d}(v, v') = F_{\hat{\theta}}(\pi(v \text{ at } p, v' \text{ at } p+d)) - u_c(v) - u_c(v') - F_{\hat{\theta}}(\pi(r)), \quad (22)$$

for the coarse adjacencies d (east, south, the two diagonals). Exact for a linear F_θ . For large D_c , skip the tables and evaluate $F_{\hat{\theta}}$ on the candidate's painted footprint at sampling time. Upward recursion: for the next coarser channel, c with $E_c + \Phi_c$ is the fine level, and the AIS base p_0 is built from c 's unaries (designed and learned u_c).

Gauge. Only y is observed, so u and $\{g_\delta\}$ are identified up to the transformations that leave every window sum invariant (e.g. moving a constant from u into g_δ). The materialised coarse tables are invariant and are what is used. When comparing coarse tables between methods, fix the reference gauge ($g_c(r, \cdot) = g_c(\cdot, r) = 0$) or double-centre (subtract row and column means).

5 The conv potential (coarse-level pair potentials)

A parametric factor over one channel, kept for coarse levels. Each value has a feature vector $e : \mathcal{D} \rightarrow \mathbb{R}^k$, the channel’s *feature set*, fixed before fitting: the rows of a free matrix $M \in \mathbb{R}^{D \times k}$ (one-hot, or learned like token embeddings), or derived from what the value asks of the finer level (the *stamp*: paint v alone in its block’s frame and take the indicator over footprint cells and paint values). Offsets $d \in \{-1, 0, 1\}^2$ index the 3×3 neighbourhood. Then

$$E(z) = \sum_p \left[a \cdot e(z_p) + \sum_{d \neq 0} e(z_p)^\top A_d e(z_{p+d}) + v \cdot \text{softplus} \left(\sum_d W_d e(z_{p+d}) + b \right) \right], \quad (23)$$

with $a \in \mathbb{R}^k$, $A_d \in \mathbb{R}^{k \times k}$, $W_d \in \mathbb{R}^{m \times k}$, $b, v \in \mathbb{R}^m$; off-grid neighbours use $e(\text{pad})$ or drop out. The softplus term with m hidden units (the *plaquette head*) adds 3- and 4-body interactions. With one-hot M and $m = 0$ this is the 8-offset pair-table model and the fit (same regression as above) is closed-form ridge; for $m > 0$ it is fitted by Adam. With stamp features and A_d tied to the geometry (entry (s, s') of A_d is g_δ for cell pairs within the stencil offsets $\mathcal{O}$, zero otherwise) the free parameters are the stencil $(u, \{g_\delta\})$ and the factor is the paint potential of Section 4 written over coarse values; its materialised tables are exact. Candidate cost in the kernel: $9k$ operations (bilinear) + $9km$ (head, updated incrementally), or table lookups when materialised. On the toys the head added nothing measurable; the bilinear form with one-hot or learned features reached the estimator’s noise floor at the top level but did not generalise to held-out value pairs at the mid level; stamp features did.

6 Self-play moment matching (retained as a diagnostic)

The earlier trainer. The learned tables are exponential-family parameters θ_k , one per table entry, with sufficient statistics $N_k(z)$ (how often entry k occurs in z). For a forward sample $z \sim q$ and a one-site kernel that leaves p^* invariant (the full p^* conditional at a site, including factors homed below it),

$$\Delta_k(z) = \sum_p \sum_t p^*(z_p = t \mid z_{-p}) [N_k(z \text{ with } z_p = t) - N_k(z)], \quad \theta_k \leftarrow \theta_k - \eta \mathbb{E}_q[\Delta_k], \quad (24)$$

the Rao-Blackwellised expected change of the statistics under one step of that kernel. Contrastive divergence (CD- K) replaces the bracket by $N_k(z_K) - N_k(z) + \Delta_k(z_K)$ after K sweeps of the p^* kernel. Fixed point: q invariant under the kernel on the features. Under hard honour the p^* conditional of a coarse site is a point mass (its children pin it), so $\Delta \equiv 0$ and nothing is learned; this is why the free-energy route replaced it.

7 The toys’ exact quantities

Circles. Tiles are air or dirt, with a field μ (unary cost μ per air tile). The level above (*objects*, blocks of 8) holds either *absent* or a disc at one of 16 offsets inside the block; a present object demands dirt on its 13-cell disc and air on the surrounding ring, at a soft-honour cost κ per violated tile. The painter writes each tile’s *demand* $\pi_t \in \{\text{free, dirt, air, conflict}\}$ (conflict: in one object’s disc and another’s ring). Given the objects, tiles are independent, with per-tile partition functions

$$Z_{\text{free}} = 1 + e^{-\mu}, \quad Z_{\text{dirt}} = 1 + e^{-\mu - \kappa}, \quad Z_{\text{air}, n} = e^{-\mu} + e^{-n\kappa}, \quad Z_{\text{conflict}} = e^{-\kappa} + e^{-\mu - \kappa}, \quad (25)$$

where n is the number of rings covering an air-demand tile. So $F(\text{objects}) = -\sum_t \log Z_{\pi_t}$; a conflict tile costs $-\log Z_{\text{conflict}} + \log Z_{\text{dirt}} + \log Z_{\text{air},1} - \log Z_{\text{free}}$ and a ring tile shared by two objects gains $-\log Z_{\text{air},2} + 2\log Z_{\text{air},1} - \log Z_{\text{free}}$, relative to the objects placed separately. The induced potential on the object level is exactly a unary paint potential ($g_\delta \equiv 0$). An isolated object costs $F_{\text{obj}} = F(\text{one object}) - F(\text{none})$; the designed *presence bonus* (unary $-b$ on every present value) with $b = F_{\text{obj}} - \log 16$ makes an isolated block present or absent with equal probability.

Potts. Tiles carry q colours with cyclic distance $\rho(c, c') = \min(|c - c'|, q - |c - c'|)$ and a neighbour coupling $J\rho$; the level above (*mid*, blocks of B) holds a block colour m with honour cost $\kappa \mathbb{1}[c \neq m]$ per tile; the top level holds a palette (a pair of adjacent colours) with honour cost λ per mid cell outside it. At $\kappa \rightarrow \infty$ the induced coupling between adjacent blocks is $g(m, m') = B J \rho(m, m')$ per shared edge of length B ; the mid level is then a q -state clock model with coupling BJ , ordered for $J \gtrsim 0.2$ at $B = 4, q = 4$. Block free energies are exact by a column transfer matrix over q^B states.